REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

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1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

Metric	Value
Total Network Flows	8400
Normal Flows	7917
Detected Attacks	483
Attack Rate	5.75%

2. Machine Learning Model

Parameter	Value
Algorithm	Random Forest Classifier
Training Features	duration, protocol_type, src_bytes, dst_bytes, count
Classification	Normal / Attack
Optimized Threshold	0.50
Validation Accuracy	0.76%
Attack Precision	0.96
Attack Recall	0.60
Attack F1-Score	0.74

3. Network Protocol Distribution

Protocol	Flow Count
TCP	3785
UDP	3082
HTTPS	1109
QUIC_OR_UDP443	272
DNS	145
HTTP	3
OTHER	2
ICMP	1
DHCP	1

4. Detected Attack History

Time	Source	Destination	Protocol	Confidence
2026-08-26 19:00:37	192.168.1.18:54704	172.64.155.209:443	HTTPS	75.58%
2026-08-26 17:22:56	192.168.1.18:50768	172.64.155.209:443	HTTPS	65.62%
2026-08-26 17:21:37	192.168.1.18:50768	172.64.155.209:443	HTTPS	58.68%
2026-08-26 17:21:24	192.168.1.18:53034	140.82.113.21:443	HTTPS	77.95%
2026-08-26 16:04:17	192.168.1.18:64569	104.18.32.47:443	HTTPS	61.83%
2026-08-26 16:03:44	192.168.1.18:64569	104.18.32.47:443	HTTPS	73.3%
2026-08-26 15:53:10	192.168.1.18:53195	104.18.32.47:443	HTTPS	64.63%
2026-08-26 15:52:59	192.168.1.18:56678	140.82.114.21:443	HTTPS	55.11%
2026-08-26 15:52:47	192.168.1.18:57424	4.150.223.109:443	HTTPS	68.91%
2026-08-26 15:52:44	192.168.1.18:54768	20.199.39.224:443	HTTPS	59.56%
2026-08-26 15:50:48	0.0.0.0:68	255.255.255.255:67	DHCP	57.79%
2026-08-26 15:48:06	192.168.1.18:61716	140.82.112.22:443	HTTPS	59.51%
2026-08-26 15:41:03	192.168.1.18:64641	140.82.114.22:443	HTTPS	56.35%
2026-08-26 14:22:36	192.168.1.18:64332	104.18.32.47:443	HTTPS	51.52%
2026-08-26 14:22:20	4.2.2.2:53	192.168.1.18:58058	DNS	84.42%
2026-08-26 14:22:20	4.2.2.2:53	192.168.1.18:51135	DNS	93.99%
2026-08-26 14:22:11	192.168.1.18:59098	52.37.131.56:443	HTTPS	53.97%
2026-08-26 14:22:11	192.168.1.18:50807	3.111.241.224:443	HTTPS	50.75%
2026-08-26 14:21:38	192.168.1.18:51639	54.187.159.182:443	HTTPS	54.15%
2026-08-26 14:20:52	192.168.1.18:64332	104.18.32.47:443	HTTPS	61.27%

5. Security Recommendations

1. Investigate repeated attack predictions from the same source address.

2. Monitor network flows with unusually high traffic volume or packet counts.

3. Review suspicious TCP and UDP connections identified by the ML classifier.

4. Use the attack confidence score together with network context when validating alerts.

5. Continue collecting representative network traffic to improve the IDS model.

6. Periodically retrain and evaluate the model using updated datasets.

6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.